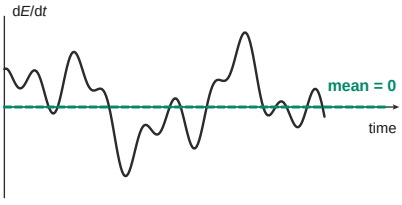
**a** **Energy balance** of a stationary state

each snapshot: not zero




**Per-sample constraint**: not what the law says



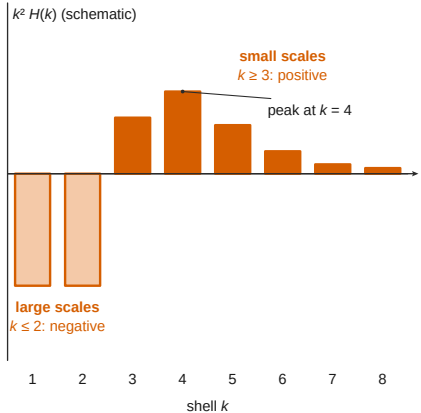

**Identity of the stationary law**

$$\mathrm{E}[\ \langle f\cdot u\rangle - \nu\langle|\omega|^2\rangle - \eta\langle|j|^2\rangle\ ] = 0$$

forcing · viscous, $u$ · Joule, $B$

residual below $4\times10^{-4}$ on the reference

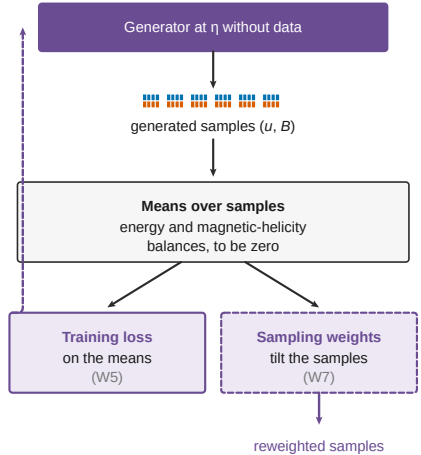
**b** **Magnetic helicity** by shell




$\Sigma_k\, k^2\, H(k) = \mathrm{E}[\langle j\cdot B\rangle] = 0$

zero mean = cancellation between scales

**c** **Use of the identities**




**Measured** (5 seeds, GPU)

W5: energy residual 0.29 → 0.19